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**Automated Diabetic Retinopathy Diagnosis via Fundus Image
Enhancement and CNN Classification**

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NORMATIVE REFERENCES

This thesis uses references to the following standards:

Instructions for the preparation of a thesis and an abstract, Higher Attestation Commission of the Ministry of Education and Science of the Republic of Kazakhstan dated September 28, 2004 No. 377-3y.

GOST 7.32-2001. Report on research work. Structure and design rules.

GOST 7.1-2003. Bibliographic record. Bibliographic description. General requirements and rules of compilation.

ST RK 34.005-2002. Information Technology. Basic terms and definitions (first edition).

ST RK 34.015-2002. Information Technology. Set of standards for automated systems. Terms of reference for creating an IS (first edition).

ST RK 34.027-2006. Information Technologies. Classification of software tools (first edition).

ST RK 34.014-2002. Information Technology. A set of standards for automated systems. Automated systems. Terms and definitions.

DESIGNATIONS AND ABBREVIATIONS

AI	Artificial Intelligence
ALO	Attention–Lesion Overlap (explainability metric)
APTOS	Asia Pacific Tele-Ophthalmology Society (2019 Blindness Detection dataset)
AUC	Area Under the Curve
BYOL	Bootstrap Your Own Latent (self-supervised learning method)
CAM	Class Activation Mapping
CLAHE	Contrast-Limited Adaptive Histogram Equalization
CNN	Convolutional Neural Network
CNR	Contrast-to-Noise Ratio
DDR	Diabetic Retinopathy grading dataset (DDR)
DINO	self-Distillation with No labels (self-supervised learning method)
DR	Diabetic Retinopathy
ECE	Expected Calibration Error
EHR	Electronic Health Record
EMD	Earth Mover's Distance
EyePACS	Eye Picture Archive Communication System (diabetic-retinopathy dataset)
FOV	Field of View
GDPR	General Data Protection Regulation
Grad-CAM	Gradient-weighted Class Activation Mapping
HIPAA	Health Insurance Portability and Accountability Act
IDRiD	Indian Diabetic Retinopathy Image Dataset
IoU	Intersection over Union
KL	Kullback–Leibler divergence
LAB	CIELAB colour space
MLP	Multilayer Perceptron
MoCo	Momentum Contrast (self-supervised learning method)
NPV	Negative Predictive Value
OD	Optic Disc
ODIR-5K	Ocular Disease Intelligent Recognition dataset (5,000 patients)
PACS	Picture Archiving and Communication System
PPV	Positive Predictive Value
ReLU	Rectified Linear Unit
RFMiD	Retinal Fundus Multi-disease Image Dataset
RIQA	Retinal Image Quality Assessment
ROC	Receiver Operating Characteristic
SimCLR	Simple framework for Contrastive Learning of Representations
SSIM	Structural Similarity Index
SSL	Self-Supervised Learning
STARE	Structured Analysis of the Retina (dataset)
UML	Unified Modeling Language
VRAM	Video Random-Access Memory

VVI	Vessel Visibility Index
WSL	Windows Subsystem for Linux
XAI	Explainable Artificial Intelligence
D	Field-of-view diameter, in pixels
CL	Clip limit of the CLAHE transform
F1	F1-score (harmonic mean of precision and recall)
G	Generalization ratio, $G = F1_external / F1_EyePACS$
T	Global threshold of the dual-constraint CLAHE (T/80 formulation)
α	Class-weight parameter of the Focal Loss (inverse-frequency)
γ	Focusing parameter of the Focal Loss ($\gamma = 2$)
κ	Cohen's Kappa with quadratic weights
σ	Standard deviation of the Gaussian kernel in flat-field correction, $\sigma = 0.07 \cdot D$
$\Delta F1$	Cross-dataset F1 drop, $\Delta F1 = F1_EyePACS - F1_external$

DEFINITIONS

Architectural complexity — the capacity of a CNN as defined by the number of convolutional layers, total trainable-parameter count, filter-size range, and the presence or absence of regularization components (batch normalization, dropout).

Baseline (baseline arm) — the reference configuration of Experiment 1: stretch-resize to 512×512 followed by ImageNet normalization, producing a 3-channel tensor with no field-of-view mask and no preprocessing stages; the backbone is initialized from ImageNet.

CLAHE (Contrast-Limited Adaptive Histogram Equalization) — an adaptive contrast-enhancement method operating on local image tiles with a clip limit that bounds contrast amplification; applied in the pipeline to the LAB L-channel under a dual-constraint clip limit (Stage 5).

Composite independent variable — the combined manipulated factor of Experiment 1 (H-1), in which the full-pipeline arm differs from baseline jointly along the preprocessing axis (eight-stage pipeline vs. stretch-resize + ImageNet normalize) and the pretraining axis (ophthalmology-specific self-supervised pretraining vs. ImageNet).

Convolutional Neural Network (CNN) — a neural architecture comprising convolutional and pooling layers for feature extraction and fully connected layers for classification; the classification backbone operating on preprocessed fundus images for five-class DR staging.

Cross-validation — a validation strategy using 5-fold patient-level stratified splitting; for each fold, four folds serve as training data and one as test data, with no patient's images appearing in both partitions; metrics are reported as mean $\pm$ standard deviation across folds.

Data augmentation — image transformations (unified affine, ColorJitter [brightness/contrast/saturation/hue], Gaussian noise, JPEG compression) applied to the training data to increase variability and improve generalization; the train-only Stage 6 of the pipeline.

Dataset-specific normalization — channel-wise normalization (Stage 7) using mean and standard deviation computed from the training split of the primary dataset rather than ImageNet statistics; the normalization of the full pipeline.

Diabetic retinopathy (DR) — a microvascular complication of diabetes mellitus and a leading cause of preventable vision loss, graded in five severity stages (0 — none, 1 — mild, 2 — moderate, 3 — severe, 4 — proliferative).

Diagnostic effectiveness — the joint performance profile on four primary metrics (accuracy, weighted F1-score, ROC-AUC, and Cohen's Kappa with quadratic weights) computed on the held-out test partition.

Domain shift — the distributional difference between training data and target deployment data arising from differences in imaging equipment, patient populations, or acquisition protocols.

Field-of-view (FOV) mask — a binary spatial mask (1.0 = real fundus data, 0.0 = zero-padding) generated at Stage 3 and appended as the 4th input channel, allowing the CNN to distinguish genuine retinal pixels from padded background.

Flat-field correction (adaptive) — Gaussian-blur subtraction with adaptive $\sigma = 0.07 \cdot D$ (D = FOV diameter), applied inside the FOV mask to normalize uneven illumination while preserving local contrast; Stage 4 of the pipeline.

Fundus image — a retinal photograph acquired by fundoscopy; the exclusive imaging modality of the dissertation and the input to the preprocessing pipeline and the CNN classifier.

Generalization (cross-database) — the ratio of test-set F1-score on a secondary dataset to test-set F1-score on the primary dataset under the same trained model without retraining; the generalization ratio $G = F1_external / F1_EyePACS$.

Grad-CAM (Gradient-weighted Class Activation Mapping) — an explainability method producing class-discriminative localization maps from a gradient-weighted combination of final convolutional feature maps; an interpretability tool, not a clinical localization of pathology.

Image quality — the measurable capacity of a fundus image to support automated detection of microvascular features relevant to DR staging, assessed through downstream classification performance rather than a subjective visual score.

Integrated pipeline dominance (H-1) — the primary hypothesis that the integrated full-pipeline configuration outperforms the baseline as a unitary system, validated when the empirical dominance criterion (≥ 5 pp weighted-F1 gain, ≥ 0.02 ROC-AUC gain, no Cohen's Kappa degradation) is satisfied.

Ophthalmology-specific self-supervised pretraining — self-supervised pretraining of the CNN backbone on an unlabeled retinal fundus corpus (no DR labels), used as the full-pipeline-arm initialization; learns fundus-domain representations directly on the 4-channel pipeline tensor.

Physician-in-the-loop — a design paradigm in which the clinician acts not merely as a consumer of AI outputs but as a tuner and auditor who interprets results and adjusts the system; the proposed system is a decision-support tool within this paradigm.

Preprocessing pipeline — the canonical ordered eight-stage sequence (Stages 0–7) applied to fundus images prior to CNN input; the full pipeline applies all eight stages, producing a 4-channel tensor (RGB + FOV mask).

Resource-limited environment — a deployment context characterized by at least two of: absence of GPU acceleration; available RAM below 16 GB; batch-processing-time constraints; network connectivity precluding continuous cloud-API reliance.

Transfer learning — the reuse and adaptation of pretrained network weights to the fundus-image domain; under H-1 the initialization source is slaved to the arm (ImageNet for baseline, ophthalmology-specific SSL for the full pipeline).

Weighted loss function — a cross-entropy loss with class-specific weights addressing class imbalance and exploiting the ordinal structure of the five-class DR grading.